

Gesesew Reta Habtie

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Employment

- 2025 – **Head of Department** of Physics, Debre Berhan University, DB, Ethiopia.
- 2024 – **Assi. Prof.** Department of Physics, Debre Berhan University, DB, Ethiopia.
- 2016 – 2019 **Lecturer.** Department of Physics, Debre Berhan University, DB, Ethiopia.
- 2012 – 2016 **Lecturer.** Department of Physics, Bulehora University, BH, Ethiopia.

Education

- 2025 **Ph.D. University of Calcutta, India.** in Astrophysics.
Thesis title: *Study of Novae.*
- 2015 **M.Sc. University of Ghana, Ghana.** in Mathematical Science.
Thesis title: *The Stationary State of Magnetic field Near the Sun.*
- 2011 **M.Sc. Dilla University, Ethiopia.** in Physics.
Thesis title: *Radiation Therapy.*
- 2009 **B.Ed. Debre Berhan University, Ethiopia.** in Physics.
Thesis title: *Attitude of Students Towards Physics.*

Skills

- Proposals & Observations I have successfully submitted and been allotted over 80 hours of observation time on the 2 m Himalayan Chandra Telescope (HCT). During this period, I observed various types of novae—including dwarf and symbiotic novae—using the **HFOC** (Gr7&8; *BVRI* filters), **TIRSPEC** (*JHK* filters), and **HESP** ($R = 30,000$) instruments.
- Data Reduction IRAF, PyRAF, PYTHON, ASTROPY.
- Programing Languages PYTHON, FORTRAN, AWK, and SHELL SCRIPTING.
- Modelling CLOUDY, SHAPE, FITYK, and PYCLOUDY.
- Operating System Linux, Mac, and MS. Window.
- Typesetting tools Latex, MS Office (word & excel), html.
- Languages Amharic (**Native**), English (**Fluent**), and Ge'ez (**Intermediate**).

Talk & Poster Presentations

- Jan. 2026 College Seminar, 2026, Debre Berhan University, Ethiopia (**Talk**).
- Oct. 2025 11th Meeting of **BRICS Astronomy Working Group (BAWG) 2025**, Brazil (**Poster**).
- Dec. 2024 College Seminar, 2024, Debre Berhan University, Ethiopia (**Talk**).
- June 2024 Symbiotic stars, weird novae, and related embarrassing binaries conference, Charles University, Prague (**Talk**).
- Feb. 2024 SNBNCBS Bose fest, SNBNCBS, Kolkata, India (**Talk**).
- Jan. 2023 42nd annual meeting of the Astronomical Society of India, IISc, Bengaluru, India (**Talk**).
- Mar. 2022 40th annual meeting of the Astronomical Society of India, IIT Roorkee, Roorkee, India (**Poster**).

Workshops and Summer School attended

- Oct. 2025  Virtual Observatory Workshop on the **BRICS Astronomy Working Group (BAWG) 2025**, Sao Paulo, Brazil.
- Apr. 2025  20th General Assembly of ESSS, Addis Ababa, Ethiopia.
- Aug. 2024  32nd IAU General Assembly, Cape Town, South Africa.
- Jan. 2024  4th Meeting on Star Formation Star Formation Studies in India, SNBNCBS, Kolkata, India.
- Jan. 2023  UV-Optical-IR Astronomy in India: Prospects of the next decade, IISc, Bengaluru, India.
- Mar. 2018  40th International School for Young Astronomers, Cairo university, Egypt.
- June 2015  Computing in Applied Mathematics (CAMS) workshop, AIMS Ghana, Ghana.

Awards

- 2025  Selected for the C2F Postdoctoral Fellowship (offer not taken).
- 2019  Selected for the TWAS-Bose Ph.D. Fellowship.
- 2017  Awarded second prize in NASA's CCMC Student Research Contest.
- 2014  Selected for the AIMS Ghana M.Sc. Fellowship.

Research Grant

- 2019  won 30,000 ETB to conduct a research on "*Investigation of Indoor Radon Concentration in the Dwelling areas of Debre Berhan Town, Ethiopia*".
- 2026  won 380,000 ETB to conduct a research on "*Documenting and Investigating Indigenous Knowledge of Calendars, Astronomy, and Prediction in North Shoa Zone, Ethiopia*".

Teaching

- 2025 –  Space Science. Debre Berhan University, DB, Ethiopia.
- 2024 – 2025  General Physics. Debre Berhan University, DB, Ethiopia.
- 2016 – 2019  Introduction to Astronomy, Stellar Physics, Nuclear Physics, Electrodynamics, Mechanics and Heat, Electromagnetism, and Modern Physics. Debre Berhan University, DB, Ethiopia.
- 2012 – 2014  Mathematical Methods of Physics, Medical Physics, Radiation Physics, Mechanics and Heat, Electromagnetism, and Modern Physics. Bule Hora University, BH, Ethiopia.

References

1. **Dr. Ramkrishna Das**, SNBNCBS, JD Block, Sector III, Salt Lake, 700106, Kolkata, India.
Email: ramkrishna.das@bose.res.in
2. **Dr. Elias Aydi**, Texas Tech University, Box 41051, Lubbock, TX, 79409-1051, USA.
Email: eydi@ttu.edu
3. **Prof. N. M. Ashok**, PRL, Navrangpura, Ahmedabad 380 009, Gujarat, India.
Email: ashoknagarhalli@gmail.com
4. **Dr. Tapas Baug**, SNBNCBS, JD Block, Sector III, Salt Lake, 700106, Kolkata, India.
Email: tapasbaug@bose.res.in

Research Publications

Refereed Journal Articles

- 1 E. Aydi, J. D. Monnier, A. Mérand, *et al.*, “Multiple outflows and delayed ejections revealed by early imaging of novae,” *Nature Astronomy*, 2025.  DOI: 10.1038/s41550-025-02725-1.
- 2 **G. R. Habtie** and R. Das, “Optical spectroscopy and temporal evolution of the nova v1405 cas,” *The Astrophysical Journal*, 2025, under review.
- 3 **G. R. Habtie** and R. Das, “Spectroscopic study of the quiescent stages in between the 2006 and 2021 outbursts of rs ophiuchi,” *Monthly Notices of the Royal Astronomical Society*, vol. 537, no. 2, pp. 2046–2060, 2025.  DOI: 10.1093/mnras/staf119. arXiv: 2402.03234 [astro-ph. SR].
- 4 **G. R. Habtie**, R. Das, R. Pandey, N. M. Ashok, and P. A. Dubovsky, “Study of the fastest classical nova, v1674 her: Photoionization and morpho-kinematic model analysis,” *Monthly Notices of the Royal Astronomical Society*, vol. 527, no. 1, pp. 1405–1423, 2024.  DOI: 10.1093/mnras/stad3295. arXiv: 2310.16619 [astro-ph. SR].
- 5 R. Pandey, **G. R. Habtie**, R. Bandyopadhyay, R. Das, F. Teyssier, and J. Guarro Fló, “Study of 2021 outburst of the recurrent nova rs ophiuchi: Photoionization and morpho-kinematic modelling,” *Monthly Notices of the Royal Astronomical Society*, vol. 515, no. 3, pp. 4655–4668, 2022.  DOI: 10.1093/mnras/stac2079.

Conference Proceedings

- 1 **G. R. Habtie**, R. Das, R. Pandey, N. M. Ashok, and P. A. Dubovsky, “Study of the fastest classical nova, v1674 her: Photoionization and morpho-kinematic model analysis,” in *42nd Meeting of the Astronomical Society of India (ASI)*, vol. 42, 2024, O51, O51.
- 2 **G. R. Habtie** and T. del Rio Gaztelurrutia, “The stationary state of magnetic field near the sun,” in *Solar Heliospheric and Interplanetary Environment (SHINE 2017)*, 2017, 170, p. 170.

Astronomer’s Telegrams / Non-Peer-Reviewed

- 1 **G. R. Habtie**, R. Das, and R. Pandey, *T crb on the verge of an outburst: H α profile evolution and accretion activity*, The Astronomer’s Telegram, No. 17041, 2025.
- 2 **G. R. Habtie** and R. Das, *He ii emission line became the most intense line in the optical spectrum of nova v1405 cas*, The Astronomer’s Telegram, No. 16496, 2024.
- 3 **G. R. Habtie** and R. Das, *Optical photometry and spectroscopy of nova sco 2024 (v1723 sco)*, The Astronomer’s Telegram, No. 16454, 2024.